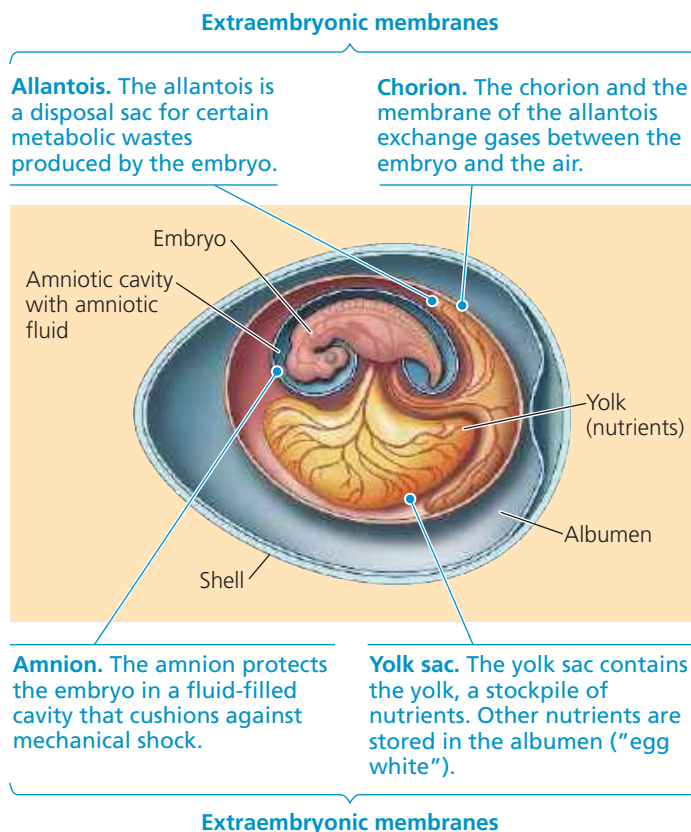


▼ **Figure 34.26 The amniotic egg.** The embryos of reptiles and mammals form four extraembryonic membranes: the allantois, chorion, amnion, and yolk sac. This diagram shows these membranes in the shelled egg of a reptile.



Most mammals have lost the eggshell over the course of their evolution, and the embryo avoids desiccation by developing within the amnion inside the mother's body.

Amniotes have acquired other key adaptations to life on land. For example, amniotes use their rib cage to ventilate their lungs. This method is more efficient than throat-based ventilation, which amphibians use as a supplement to breathing through their skin. The increased efficiency of rib cage ventilation may have allowed amniotes to abandon breathing through their skin and develop less permeable skin, thereby conserving water.

Early Amniotes

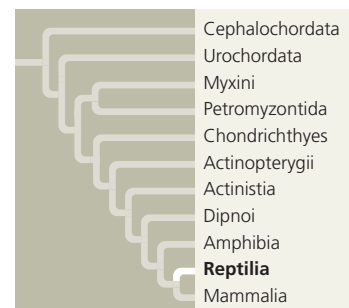
The most recent common ancestor of living amphibians and amniotes lived 350 million years ago. Based on where their fossils have been found, the earliest amniotes lived in warm, moist areas, as did the first tetrapods. Over time, early amniotes expanded into a wide range of new environments, including dry and high-latitude regions. Fossil evidence shows that the earliest amniotes resembled small lizards with sharp teeth, a sign that they were predators (**Figure 34.27**). Later groups of amniotes also included herbivores, as evidenced by their grinding teeth and other features.

▼ **Figure 34.27 Artist's reconstruction of *Hylonomus*, an early amniote.** About 25 cm long, this species lived 310 million years ago and probably ate insects and other small invertebrates.



Present-day amniotes include two large clades of terrestrial vertebrates: reptiles and mammals.

Reptiles



Living members of the **reptile** clade include turtles, tuataras, lizards and snakes, crocodilians, and birds (see **Figure 34.25**). There are about 20,800 species of reptiles, the majority of which are squamates (lizards and snakes; 10,425 species) or birds (10,000 species).

As a group, the reptiles share several derived characters that distinguish them from other tetrapods. For example, unlike amphibians, reptiles have scales that contain the protein keratin (as does a human nail). Scales help protect the animal's skin from desiccation and abrasion. In addition, most reptiles lay their shelled eggs on land; the shell protects the egg from drying out (**Figure 34.28**). Fertilization occurs internally, before the eggshell is secreted.

Reptiles such as lizards and snakes are sometimes described as "cold-blooded" because they do not use their metabolism

▼ **Figure 34.28 Hatching reptiles.** These baby panther chameleons (*Furcifer pardalis*) are breaking out of their parchment-like shells, a common type of shell among living reptiles other than birds.

